

Clinton Enwerem | Résumé

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SUMMARY

Doctoral candidate in Electrical & Computer Engineering at the University of Maryland researching robot learning, dexterous robotic grasping and manipulation, and multi-modal uncertainty estimation. Experienced in designing and implementing robust grasp planning and control algorithms for robotic arms with anthropomorphic hands, validated through high-fidelity simulation (MuJoCo, Drake) and real hardware (UR3e, xArm7, LEAP Hand). Seeking research engineering, applied ML, or robotics roles.

EDUCATION

- 2021-Date** **PhD, Electrical & Computer Engineering**, University of Maryland, College Park, MD. Advisors: Prof. John S. Baras and Prof. Calin Belta. Relevant Coursework: Decision-Making for Robotics, Nonlinear Control Systems, Optimal Control, Convex Optimization, Decision Making Under Uncertainty.
- 2014-2018** **B.Eng., Electrical Engineering**, University of Nigeria, Nsukka. First-Class Honors.

SKILLS

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| Robotics | ROS 2, MuJoCo, Graspl!, Drake |
| RL/ML | PyTorch, OpenAI Gym, Safety Gymnasium, TensorFlow, OpenCV |
| Robots/EoATs | UR3e, xArm7, LEAP Hand, RealHand L6 |
| Programming | Python, C++, MATLAB, Bash |
| Optimization | CVXPY, Gurobi, Mosek |
| Tools | Git, GitHub, Docker, LaTeX |

SELECTED PUBLICATIONS

- 2026** **Clinton Enwerem**, J. S. Baras, and C. Belta, "Risk-Constrained Belief-Space Optimization for Safe Control under Latent Uncertainty," arXiv preprint, 2026. [arXiv link](#)
- 2025** **Clinton Enwerem**, A. G. Puranic, J. S. Baras, and C. Belta, "Safety-Aware RL for Control via Risk-Sensitive Value Iteration and Quantile Regression," CDC 2025.
- 2024** **Clinton Enwerem**, E. Noorani, J. S. Baras, and B. M. Sadler, "Robust Stochastic Shortest-Path Planning via Risk-Sensitive Incremental Sampling," CDC 2024.
- 2024** **Clinton Enwerem** and J. S. Baras, "Safe Collective Control under Noisy Inputs and Competing Constraints via Non-Smooth Barrier Functions," ECC 2024.

RESEARCH EXPERIENCE

Institute for Systems Research, University of Maryland

- 8/2021-Date** **Graduate Research Assistant**
- Collaborate with PI, postdocs, and graduate researchers to design novel robust grasping and manipulation algorithms for robotic arms equipped with anthropomorphic hands.
 - Implement ROS 2-compliant software in Python and C++ for grasp planning and control.
 - Validate algorithms through large-scale simulation in MuJoCo and Drake, and RL sandboxes (OpenAI Gym, Safety Gymnasium).
 - Author conference and journal papers, technical reports, and presentations.

PROFESSIONAL EXPERIENCE

Kognitive Robotics, Lagos, Nigeria

- 3/2020-2/2021** **Robotics Engineer**
- Engineered the complete power and sensor-actuator interface circuitry for a mobile robot using Altium for schematic capture and PCB design.
 - Supported robot hardware and ROS(1)-compliant software realization for the company's MVP.

Robotics & Artificial Intelligence Nigeria (RAIN), Ibadan, Nigeria

- 3/2020-2/2021** **Robotics Trainee**
- Contributed to robotics projects across hardware and software stacks, including CAD, rapid prototyping, sensor fusion, and control firmware development.

TEACHING EXPERIENCE

ECE Department, University of Maryland

- 9/2025-12/2025** **Graduate Teaching Assistant**
- ENEE467 — Robotics Project Laboratory (Prof. Calin Belta). Mentored 50+ students in kinematics, 3D perception, motion planning, and ROS 2 control. Built reproducible Docker-based labs.

2/2025-5/2025 Graduate Teaching Fellow

- ENEE661 — Nonlinear Control Systems (Prof. John S. Baras). Led problem-solving sessions on Lyapunov stability, input-to-state stability, and nonlinear analysis.

PROJECTS

- [simple_urdf_parser](#) — Compact URDF parsing toolkit for fast manipulator kinematics and C-space sampling. (Python) [GitHub](#)
- [Imitation Learning for Grasp Stabilization](#) — Trained neural policies (MLP and LSTM) via behavior cloning from expert demonstrations to stabilize grasps under perturbations using tactile and proprioceptive observations. (Python, PyTorch, MuJoCo)

HONORS & AWARDS

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| 2024 | IEEE CSS Student Travel Award (CDC 2024). |
| 2022 | Microsoft Diversity in Robotics & Autonomy PhD Fellowship (2022-2023). |
| 2022 | ROSCON Diversity Scholarship, Kyoto, Japan. |
| 2021 | Dean's Fellowship, University of Maryland. |
| 2020 | Scholar, Stanford Exposure to Research & Graduate Education. |